



No Microchips Here

Here's the complete list of ingredients in Pfizer's COVID-19 vaccine. <http://digit.in/jan21-09>



Mr Worldwide

Antarctica reports first COVID-19 case; disease has now spread to every continent in the world. <http://digit.in/jan21-10>



FIGHTING THE PANDEMIC

PART V

The research directions to stop the spread of the novel coronavirus

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he development of vaccines is going on at an accelerated pace, with vaccines being introduced

before going through the typically longer test periods and clinical trials that would take place during more normal times. In India, state government bodies began meeting and forming the plans for roll out of the vaccine locally. There are 236 vaccines now in development with seven vaccines getting a regulatory approval from a government agency.

This is still not the final phase of vaccine development, especially since the development has been accelerated, we can expect an elongated Phase IV of the rollout as researchers continue to monitor the effects of the vaccine after it has been released to the public. Meanwhile research continues in earnest as scientists understand more about how the virus operates within the body, how it behaves in populations, and how it affects the health of communities.

DATA ON REINFECTIONS

The primary fear with reinfection is that the body is unable to develop immunity to the disease. Till date there are six documented cases of reinfection. To better understand reinfection, it is necessary to first understand a bit more about the immune system and how the tests for immunity and reinfection work.

The first line of defence are innate cells, which include phagocytes, cells that indiscriminately devour all alien material. Then there are the

specialised white blood cells known as lymphocytes, including B cells and T cells, which are flexible and reactive to diseases. The B cells form antibodies while the T cells directly attack the invaders. Then there are the memory cells, specialised lymphocytes that remember pathogens and how to deal with them. The vaccines work by directly introducing memory cells into the body, without first making it go through the dangerous infection stage.

Of course every virus is unique and tricky in its own way. There are some viruses that have evolved mechanisms to continuously outsmart the immune system, including HIV and malaria. Still others originate from animal sources, and these zoonotic viruses are entirely novel to the human body. This makes it challenging for the immune system to deal with, and SARS-CoV-2 is a zoonotic virus. While the human body does not develop robust immunity to the less harmful strains of betacoronaviruses, such as the ones